

Simple Harmonic Motion

The most important motion in physics: springs, pendulums, damping and resonance, from car suspensions to skyscrapers.

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Disturb almost any stable system slightly and it oscillates: a mass on a spring, a pendulum, a tuning fork, a bridge, a skyscraper in the wind, atoms in a crystal, the air in an organ pipe, the current in a radio circuit. Chapter 5 showed why. Near a minimum of potential energy, every smooth potential looks like a spring, $U \approx \frac{1}{2}kx^2$. The resulting motion, **simple harmonic motion** (SHM), is described by sines and cosines. It is the foundation of waves (Chapter 11), AC circuits (Chapter 21) and much of engineering vibration analysis. This chapter develops SHM, then adds the two effects that matter in practice: **damping** and **resonance**.

YOU WILL BE ABLE TO

- Recognise SHM from a linear restoring force, and write and solve its equation of motion.
- Use amplitude, angular frequency, period and phase, and match a solution to its initial conditions.
- Relate SHM to uniform circular motion, and compute velocity, acceleration and energy.
- Analyse horizontal and vertical springs, combinations of springs, simple and physical pendulums.
- Describe underdamped, critically damped and overdamped motion, and the quality factor Q .
- Explain forced oscillations and resonance, and their engineering consequences.

THINK FIRST

A grandfather clock keeps perfect time on the ground floor. Would it run fast, slow, or keep time (a) at the top of a tall mountain, and (b) in a lift accelerating upward? What about a quartz wristwatch?

A pendulum's period is $T = 2\pi\sqrt{L/g}$, so it depends on the effective g .

(a) On the mountain g is slightly smaller, so T is longer and the clock runs **slow**.

(b) In a lift accelerating upward, the effective gravity is $g + a$, so T is shorter and the clock runs **fast**.

A quartz watch uses a vibrating crystal, a mass-spring system whose frequency $\sqrt{k/m}$ does not depend on g at all. It keeps time in both places, and in orbit too.

§9.1

The equation of simple harmonic motion

A mass m attached to an ideal spring of stiffness k on a frictionless surface feels the force $F = -kx$, where x is measured from equilibrium. Newton's second law gives

$$m \frac{d^2x}{dt^2} = -kx \implies \frac{d^2x}{dt^2} + \omega^2 x = 0, \quad \omega = \sqrt{\frac{k}{m}}.$$

KEY IDEA

Definition of SHM. Any system whose displacement obeys $\ddot{x} = -\omega^2 x$ performs simple harmonic motion. The acceleration is proportional to the displacement and directed towards equilibrium. The general solution is

$$x(t) = A \cos(\omega t + \phi),$$

where the **amplitude** A and the **phase constant** ϕ are set by the initial conditions, and ω is set by the system.

To check the solution: $\dot{x} = -A\omega \sin(\omega t + \phi)$ and $\ddot{x} = -A\omega^2 \cos(\omega t + \phi) = -\omega^2 x$. ✓

Vocabulary:

- Angular frequency ω , in rad/s.
- Frequency $f = \omega/2\pi$, in hertz (Hz), meaning cycles per second.
- Period $T = 1/f = 2\pi/\omega$.
- Phase $\omega t + \phi$.

For a mass on a spring,

$$T = 2\pi\sqrt{\frac{m}{k}}.$$

The period is **independent of the amplitude**. This *isochronism* is what makes oscillators good clocks.

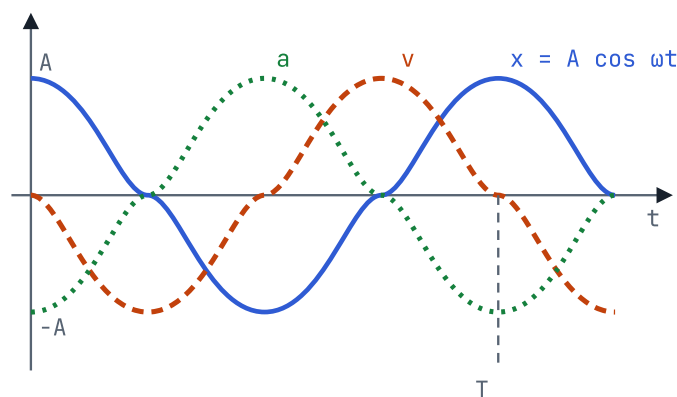
Velocity and acceleration

$$v = \frac{dx}{dt} = -A\omega \sin(\omega t + \phi), \quad a = \frac{dv}{dt} = -A\omega^2 \cos(\omega t + \phi) = -\omega^2 x.$$

- The speed is greatest, $v_{\max} = A\omega$, at equilibrium ($x = 0$).
- The acceleration is greatest, $a_{\max} = A\omega^2$, at the turning points ($x = \pm A$), where $v = 0$.
- The velocity leads the displacement by a quarter cycle (90°), and the acceleration is exactly opposite to the displacement (180° out of phase).

Eliminating t gives the useful relation

$$v = \pm\omega\sqrt{A^2 - x^2}.$$



In SHM the displacement (solid) is a cosine. The velocity (dashed) leads it by a quarter period. The acceleration (dotted) is always opposite to the displacement.

SHM and uniform circular motion

Picture a point moving round a circle of radius A at constant angular speed ω , starting at angle ϕ . Its x -coordinate is $A \cos(\omega t + \phi)$. SHM is the projection of uniform circular motion onto a diameter. This *reference circle* explains why ω is called an angular frequency. It also gives $v_{\max} = A\omega$ (the speed round the circle) and $a_{\max} = A\omega^2$ (the centripetal acceleration).

WORKED EXAMPLE 9.1

A mass on a spring

A 0.50 kg block on a frictionless surface is attached to a spring with $k = 200$ N/m. It is pulled 0.10 m from equilibrium and released from rest. Find (a) ω , f and T ; (b) the equation for $x(t)$; (c) the maximum speed and the maximum acceleration; and (d) the position and velocity at $t = 0.10$ s.

Solution

(a) $\omega = \sqrt{200/0.50} = 20$ rad/s, so $f = 20/2\pi = 3.18$ Hz and $T = 0.314$ s.

(b) It is released from rest at $x = +A$, so $\phi = 0$ and $x = 0.10 \cos(20t)$ m.

(c) $v_{\max} = A\omega = 2.0$ m/s and $a_{\max} = A\omega^2 = 40$ m/s².

(d) At $t = 0.10$ s the phase is $20(0.10) = 2.0$ rad:

$$x = 0.10 \cos 2.0 = -0.042 \text{ m}, \quad v = -2.0 \sin 2.0 = -1.82 \text{ m/s}.$$

The block is on the far side of equilibrium, still moving in the $-x$ direction.

Evaluate. Use radians for ωt , never degrees. A calculator in degree mode is the most common source of wrong answers in SHM.

WORKED EXAMPLE 9.2

Matching the initial conditions

The same oscillator ($\omega = 20$ rad/s) is at $x_0 = 0.050$ m and moving outward at $v_0 = +1.0$ m/s at $t = 0$. Find the amplitude and phase constant.

Solution

Amplitude, from $v^2 = \omega^2(A^2 - x^2)$:

$$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega}\right)^2} = \sqrt{0.0025 + 0.0025} = 0.0707 \text{ m}.$$

Phase. At $t = 0$: $x_0 = A \cos \phi$ and $v_0 = -A\omega \sin \phi$. So

$$\cos \phi = \frac{0.050}{0.0707} = 0.707, \quad \sin \phi = -\frac{1.0}{20(0.0707)} = -0.707 \Rightarrow \phi = -\frac{\pi}{4}.$$

So $x = 0.0707 \cos(20t - \pi/4)$ m.

Check: at $t = 0$, $x = 0.0707 \cos(-\pi/4) = 0.050$ m ✓ and $v = -1.414 \sin(-\pi/4) = +1.0$ m/s ✓.

§9.2

Energy in SHM

With no friction, mechanical energy is conserved:

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2 = \frac{1}{2}mv_{\max}^2.$$

Substituting $x(t)$ and $v(t)$ shows that the energy sloshes back and forth between kinetic and potential forms, twice per cycle:

$$U = \frac{1}{2}kA^2 \cos^2(\omega t + \phi), \quad K = \frac{1}{2}kA^2 \sin^2(\omega t + \phi).$$

Averaged over a cycle, $\langle K \rangle = \langle U \rangle = \frac{1}{2}E$.

WORKED EXAMPLE 9.3

Energy at half amplitude

At what displacement is the energy of an SHM oscillator shared equally between kinetic and potential? What fraction of the energy is kinetic at $x = A/2$?

Solution

Equal sharing: $\frac{1}{2}kx^2 = \frac{1}{2}(\frac{1}{2}kA^2)$, so $x = \pm A/\sqrt{2} = \pm 0.707A$.

At $x = A/2$: $U = \frac{1}{2}k(A/2)^2 = \frac{1}{4}E$, so $K = \frac{3}{4}E$. The speed is $v = \omega\sqrt{A^2 - A^2/4} = \frac{\sqrt{3}}{2}v_{\max} = 0.87v_{\max}$.

Evaluate. The oscillator spends most of its time near the turning points, where it moves slowly, and passes quickly through the middle. That is why a vibrating guitar string looks blurred at its extremes, and why the probability of finding a classical oscillator is highest near $x = \pm A$.

§9.3

Mass–spring systems

Vertical springs

A mass hanging from a spring stretches it to a new equilibrium, at which $k\Delta = mg$. Measure x from **this** equilibrium. The net force is then $-k(\Delta + x) + mg = -kx$, so the motion is the same SHM, with the same $T = 2\pi\sqrt{m/k}$. Gravity only shifts the equilibrium point. A handy result: $T = 2\pi\sqrt{\Delta/g}$, where Δ is the static stretch.

Springs in combination

- **Parallel** (side by side, sharing the load, equal extensions): $k_{\text{eff}} = k_1 + k_2$.
- **Series** (end to end, the same force through each): $\frac{1}{k_{\text{eff}}} = \frac{1}{k_1} + \frac{1}{k_2}$.

Cutting a spring in half **doubles** its stiffness, because each half stretches only half as much under the same force.

WORKED EXAMPLE 9.4

A vertical spring

A 2.0 kg mass hung from a spring stretches it by 4.9 cm. Find the spring constant and the period of vertical oscillations.

Solution

$$k = \frac{mg}{\Delta} = \frac{2.0(9.8)}{0.049} = 400 \text{ N/m}, \quad T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{2.0}{400}} = 0.44 \text{ s}.$$

$$\text{Check: } 2\pi\sqrt{\Delta/g} = 2\pi\sqrt{0.049/9.8} = 0.44 \text{ s. } \checkmark$$

§9.4

Pendulums

The simple pendulum

A point mass m on a light string of length L , displaced by angle θ , feels a restoring torque about the pivot of $-mgL \sin \theta$. With $I = mL^2$:

$$mL^2\ddot{\theta} = -mgL \sin \theta \implies \ddot{\theta} = -\frac{g}{L} \sin \theta.$$

This is **not** SHM in general, because of the $\sin \theta$. For small angles, $\sin \theta \approx \theta$ (in radians), and

$$\ddot{\theta} \approx -\frac{g}{L} \theta \Rightarrow \omega = \sqrt{\frac{g}{L}}, \quad T = 2\pi \sqrt{\frac{L}{g}}.$$

The period is independent of the mass, and, for small amplitudes, of the amplitude too. This confirms the dimensional-analysis guess of Example 1.4, with $k = 2\pi$. For larger amplitudes θ_0 , the period grows slightly:

$$T \approx T_0 \left(1 + \frac{\theta_0^2}{16} + \dots \right).$$

At $\theta_0 = 15^\circ$ this is only a 0.4% correction.

The physical pendulum

Any rigid body swinging about a pivot a distance d from its centre of mass feels a gravitational torque $-mgd \sin \theta$. For small angles:

$$I\ddot{\theta} = -mgd \theta \implies T = 2\pi \sqrt{\frac{I}{mgd}},$$

where I is the moment of inertia about the pivot. A simple pendulum is the special case $I = md^2$.

The torsion pendulum

A disc hanging from a wire that twists through an angle θ feels a restoring torque $-\kappa\theta$, where κ is the wire's torsion constant. The period is $T = 2\pi\sqrt{I/\kappa}$. Cavendish used a torsion pendulum to measure G . Mechanical watches use the same principle in their balance wheel.

WORKED EXAMPLE 9.5

A pendulum clock

(a) What length gives a simple pendulum a period of exactly 2.00 s at $g = 9.80 \text{ m/s}^2$? Such a pendulum “beats seconds”: each swing takes 1 s. (b) The clock is moved to a place where $g = 9.78 \text{ m/s}^2$. How many seconds per day does it gain or lose?

Solution

(a) Solve $T = 2\pi\sqrt{L/g}$ for L :

$$L = \frac{gT^2}{4\pi^2} = \frac{9.80(4.00)}{39.48} = 0.993 \text{ m.}$$

(b) $T \propto g^{-1/2}$, so

$$\frac{T'}{T} = \sqrt{\frac{9.80}{9.78}} = 1.00102.$$

Each swing is 0.102% longer, so the clock loses $0.00102 \times 86\,400 = 88 \text{ s}$ per day.

Evaluate. Pendulum clocks were once accurate enough to *measure* g . Variations in their rate around the world gave the first evidence that the Earth is not a perfect sphere.

WORKED EXAMPLE 9.6

Your leg as a pendulum

Model a leg as a uniform rod 0.90 m long, swinging freely from the hip. Find its natural period of swing, and use it to estimate a comfortable walking pace.

Solution

For a rod pivoted at one end, $I = \frac{1}{3}mL^2$ and $d = L/2$:

$$T = 2\pi\sqrt{\frac{\frac{1}{3}mL^2}{mg(L/2)}} = 2\pi\sqrt{\frac{2L}{3g}} = 2\pi\sqrt{\frac{2(0.90)}{3(9.8)}} = 1.55 \text{ s.}$$

A step takes about half a period, 0.78 s. With a step length of about 0.75 m, the walking speed is roughly $0.75/0.78 \approx 1 \text{ m/s}$.

Evaluate. Typical relaxed walking speeds are 1.2–1.4 m/s, so this very simple model is surprisingly close. Walking at the leg’s natural frequency minimises the effort spent swinging it. Since $T \propto \sqrt{L}$, tall people and long-legged animals walk with a slower cadence. Clinical gait analysis uses these pendulum models to separate normal from pathological patterns.

§9.5

Damped oscillations

Real oscillators lose energy to friction, air resistance or internal losses. Model this with a damping force proportional to velocity, $-b\dot{x}$, as from a dashpot or shock absorber:

$$m\ddot{x} + b\dot{x} + kx = 0.$$

Define $\omega_0 = \sqrt{k/m}$ (the natural frequency) and $\gamma = b/2m$ (the damping rate). There are three types of solution:

- **Underdamped** ($\gamma < \omega_0$): the system oscillates inside a decaying envelope,

$$x = Ae^{-\gamma t} \cos(\omega_d t + \phi), \quad \omega_d = \sqrt{\omega_0^2 - \gamma^2}.$$

- **Critically damped** ($\gamma = \omega_0$): it returns to equilibrium as fast as possible without overshooting.
 $x = (C_1 + C_2 t)e^{-\omega_0 t}.$
- **Overdamped** ($\gamma > \omega_0$): it creeps slowly back to equilibrium without oscillating.

The energy of a lightly damped oscillator decays as $E \propto e^{-2\gamma t}$. The **quality factor**

$$Q = \frac{\omega_0}{2\gamma} = \frac{m\omega_0}{b} \approx 2\pi \times \frac{\text{energy stored}}{\text{energy lost per cycle}}$$

measures how many cycles the system rings for. A car suspension has $Q \sim 1$, a guitar string $Q \sim 10^3$, a quartz crystal $Q \sim 10^5$, and an atomic clock transition $Q \sim 10^{15}$.

WORKED EXAMPLE 9.7

Damping a car suspension

One wheel of a car carries 350 kg (a quarter of the car's mass) on a spring with $k = 35\,000$ N/m. (a) Find the natural frequency. (b) Find the damping coefficient for critical damping. (c) Engineers typically choose about 30% of critical damping. Why not 100%?

Solution

(a) $\omega_0 = \sqrt{35\,000/350} = 10$ rad/s, so $f_0 = 1.6$ Hz.

(b) Critical damping requires $\gamma = \omega_0$, that is, $b_c = 2m\omega_0 = 2\sqrt{km}$:

$$b_c = 2\sqrt{35\,000 \times 350} = 2\sqrt{1.225 \times 10^7} = 7.0 \times 10^3 \text{ N s/m}.$$

(c) **Trade-off.** Critical damping gives the fastest return *without overshoot*. But a stiff damper also transmits more of the road's bumps straight to the car body. With about 30% of critical ($b \approx 2100 \text{ N s/m}$), the car overshoots slightly, then settles within a cycle or so, giving a smoother ride. Shock absorbers have no shock-absorbing role as such. They are *dampers* that stop the springs from oscillating.

§9.6

Forced oscillations and resonance

Drive a damped oscillator with a periodic force $F_0 \cos \omega t$:

$$m\ddot{x} + b\dot{x} + kx = F_0 \cos \omega t.$$

After the starting transients die away, the system oscillates at the **driving** frequency ω , not at its natural frequency, with steady-state amplitude

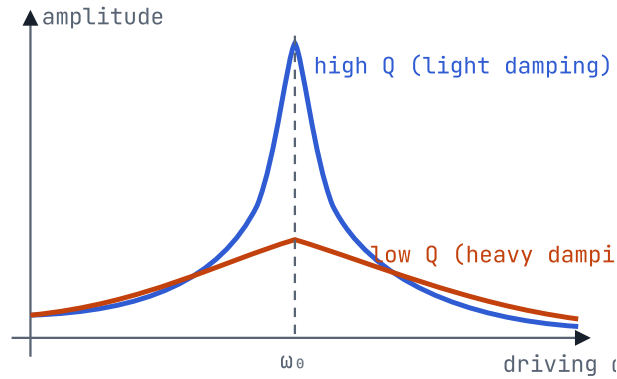
$$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}}.$$

KEY IDEA

Resonance. The amplitude is largest when the driving frequency is close to the natural frequency, $\omega \approx \omega_0$. With light damping, the peak amplitude is

$$A_{\max} \approx \frac{F_0}{b\omega_0} = Q \frac{F_0}{k},$$

which is Q times the static deflection F_0/k . The width of the resonance peak (between the half-power points) is $\Delta\omega = 2\gamma = \omega_0/Q$. A high- Q system responds enormously, but only within a very narrow band of frequencies.



Resonance. With light damping the response peaks sharply at the natural frequency. Heavy damping lowers and broadens the peak.

WORKED EXAMPLE 9.8

A driven oscillator

A 2.0 kg mass on a spring ($k = 800$ N/m) with damping $b = 4.0$ N s/m is driven by a force of amplitude 10 N. Find (a) ω_0 and Q , (b) the static deflection and the resonant amplitude, and (c) the amplitude when driven at $\omega = 10$ rad/s.

Solution

(a) $\omega_0 = \sqrt{800/2.0} = 20$ rad/s and $\gamma = b/2m = 1.0$ s⁻¹, so $Q = \omega_0/2\gamma = 10$.

(b) The static deflection is $F_0/k = 10/800 = 1.25$ cm. The resonant amplitude is Q times larger, about 12.5 cm.

(c) At $\omega = 10$ rad/s:

$$A = \frac{10/2.0}{\sqrt{(400 - 100)^2 + (2 \times 1.0 \times 10)^2}} = \frac{5.0}{\sqrt{90\,000 + 400}} = \frac{5.0}{300.7} = 1.66 \text{ cm.}$$

Away from resonance, the response is only slightly larger than the static deflection.

IN PRACTICE

Resonance: friend and foe.

- **Structures.** Every building has natural frequencies. A rough rule of thumb gives a period of about $0.1N$ seconds for an N -storey building. Earthquake engineers make sure these frequencies avoid the dominant frequencies of ground shaking at the site. The 660-tonne pendulum hanging near the top of Taipei 101 is a *tuned mass damper*: tuned to the tower's sway frequency, it swings out of phase with the building and soaks up energy from wind and earthquakes. London's Millennium Bridge had to be closed two days after opening in 2000. Pedestrians unconsciously synchronised their steps with its sideways sway, driving it at resonance, and dampers had to be retrofitted.
- **Machines.** Rotating machinery (turbines, washing machines, car engines) must pass quickly through, or stay away from, the speeds that excite structural resonances. Engine mounts and vibration isolators are mass-spring-damper systems designed for this.
- **Medicine and the senses.** Inside the cochlea, the basilar membrane is stiff near its base and floppy near its tip, so each place along it resonates at a different frequency. This *tonotopic* map is how the ear separates pitch. Ultrasound transducers are piezoelectric crystals driven at their resonant frequency (2–15 MHz). MRI works by driving hydrogen nuclei at their magnetic resonance frequency (Chapter 18).
- **Timekeeping and sensing.** A quartz watch counts the oscillations of a crystal tuned to 32 768 Hz (2^{15}). MEMS accelerometers and the cantilevers of atomic force microscopes are tiny resonators whose shifts in frequency reveal forces and masses.

CHAPTER SUMMARY

- SHM: $\ddot{x} = -\omega^2 x$, with solution $x = A \cos(\omega t + \phi)$, $f = \omega/2\pi$ and $T = 2\pi/\omega$.
- $v_{\max} = A\omega$ at equilibrium. $a_{\max} = A\omega^2$ at the turning points. $v = \pm\omega\sqrt{A^2 - x^2}$.
- SHM is the projection of uniform circular motion. Energy $E = \frac{1}{2}kA^2$ is shared between K and U .
- Mass-spring: $T = 2\pi\sqrt{m/k}$, the same for vertical springs. Parallel: $k_1 + k_2$. Series: $(1/k_1 + 1/k_2)^{-1}$.
- Simple pendulum: $T = 2\pi\sqrt{L/g}$ (small angles). Physical pendulum: $T = 2\pi\sqrt{I/mgd}$. Torsion pendulum: $T = 2\pi\sqrt{I/\kappa}$.
- Damping: $x = Ae^{-\gamma t} \cos \omega_d t$, with $\gamma = b/2m$ and $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$. Critical damping at $\gamma = \omega_0$. $Q = \omega_0/2\gamma$.
- Forced oscillations: the response peaks near ω_0 with $A_{\max} \approx QF_0/k$, and the peak width is ω_0/Q .

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P9.1

A ball bounces repeatedly on a hard floor, returning to the same height each time. Is this simple harmonic motion? Explain using the definition of SHM.

P9.2

A mass-spring oscillator and a simple pendulum both have a period of 1.0 s on Earth. What are their periods on the Moon, where g is about one-sixth of its Earth value?

P9.3

In SHM, where is the speed greatest? Where is the magnitude of the acceleration greatest? Can the velocity and acceleration ever point in the same direction?

P9.4

Why are soldiers ordered to “break step” when marching across a bridge? Explain using the idea of resonance.

LEVEL B – STANDARD PROBLEMS**P9.5**

A particle moves according to $x = 0.040 \cos(5\pi t + \pi/3)$, with x in metres and t in seconds. Find (a) the amplitude, frequency, period and phase constant, and (b) its position, velocity and acceleration at $t = 0.10$ s.

P9.6

A 0.50 kg mass on a spring with $k = 50$ N/m oscillates with an amplitude of 8.0 cm. Find (a) the total energy, (b) the maximum speed, and (c) the speed when the displacement is 4.0 cm.

P9.7

Springs with $k_1 = 100$ N/m and $k_2 = 300$ N/m are used to support a 1.0 kg mass. Find the period of oscillation when the springs are connected (a) in parallel and (b) in series.

P9.8

Hanging a 0.20 kg mass from the lower end of a light vertical spring stretches the spring by 4.0 cm. The mass is then pulled down a further 3.0 cm and released. Find the period and the maximum speed.

P9.9

An astronaut on the Moon ($g = 1.62 \text{ m/s}^2$) times a 1.0 m pendulum. What period does she measure? How would she use this to measure g on an unknown planet?

P9.10

A uniform disc of radius 0.20 m swings in its own plane about a horizontal axis through a point on its rim. Find the period of small oscillations, and the length of a simple pendulum with the same period.

P9.11

The amplitude of a damped oscillator with a period of 0.50 s falls to half its initial value after 10 complete oscillations. Find (a) the damping rate γ , (b) the quality factor Q , and (c) the fraction of the initial energy remaining after these 10 oscillations.

P9.12

Suspension and speed bumps. A quarter-car (350 kg on a spring with $k = 35\,000$ N/m) drives over a series of speed bumps 10 m apart. (a) At what road speed will the bumps drive the suspension at resonance? (b) Explain why such a road feels worst at one particular speed. (c) How does good damping help?

LEVEL C – CHALLENGE PROBLEMS**P9.13**

A particle of mass m moves in one dimension with potential energy $U(x) = U_0 \left(\frac{a}{x} + \frac{x}{a} \right)$ for $x > 0$, where U_0 and a are positive constants. (a) Find the equilibrium position and show that it is stable. (b) Find the angular frequency of small oscillations about it.

P9.14

A liquid of density ρ fills a U-tube of uniform cross-section A . The total length of the liquid column is L . The liquid is displaced so that one side rises by x and the other falls by x , and is then released. Ignoring friction, show that the motion is SHM with $\omega = \sqrt{2g/L}$. (Hint: use energy, or find the net force on the whole column.)

P9.15

A pendulum clock is regulated to keep perfect time when its pendulum swings with an amplitude of 5.0° . Over time the amplitude grows to 10° . Using $T \approx T_0(1 + \theta_0^2/16)$, find how many seconds per day the clock now gains or loses.

P9.16

Resonance curve. For the driven damped oscillator $m\ddot{x} + b\dot{x} + kx = F_0 \cos \omega t$: (a) substitute a trial solution $x = A \cos(\omega t - \delta)$ to derive the amplitude formula given in the notes, and find $\tan \delta$; (b) show that the amplitude peaks at $\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}$; (c) show that at $\omega = \omega_0$ the displacement lags the driving force by 90° , so that the velocity is in phase with the force and the power input is greatest.

P9.17

A 1.0 kg block rests on a frictionless surface, attached to a horizontal spring with $k = 100$ N/m, at its equilibrium position. A 0.25 kg lump of putty moving at 5.0 m/s along the spring's axis hits the block and sticks. Find (a) the velocity just after the collision, (b) the amplitude and period of the resulting oscillation, and (c) the fraction of the putty's kinetic energy lost in the collision.

P9.18

Scaling of walking. Using the leg-as-pendulum model of Example 9.6: (a) show that the natural walking cadence scales as $1/\sqrt{L}$ and the natural walking speed as $\sqrt{gL}$, assuming step length is proportional to leg length. (b) A child's legs are half as long as an adult's. Compare their natural walking speeds. (c) On the Moon, what would happen to the natural walking speed? Use this to explain why Apollo astronauts preferred to hop.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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